Example 1: Multiple Linear Regression

STAT 2509: Introduction to Statistical Modeling II

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Example 1

Consider the **MLR model**:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

with $n = 15$ observations. The data for x_1, x_2, x_3, and y are given below:

x_1	15	22	18	30	12	16	28	21	10	19	35	11
x_2	110	115	130	105	155	140	112	120	165	125	100	160
x_3	2.5	2.8	3.1	2.0	4.0	3.5	2.2	2.9	4.5	3.0	1.8	4.2
y	22.5	28.1	20.4	35.0	15.2	18.9	31.3	26.7	12.1	24.5	38.2	14.8

The R output for the above model based on the dataset is given on the next page.

Comparing all methods on mtcars

```
Call:
lm(formula = y ~ x1 + x2 + x3, data = df)

Residuals:
Min       1Q   Median       3Q      Max
-1.9090 -0.3987  0.1369  0.4424  1.2665

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 26.61393    4.83066   5.509 0.000567 ***
x1           0.81677    0.10538   7.750 5.48e-05 ***
x2          -0.22854    0.09395  -2.433 0.041032 *
x3           3.45283    2.92562   1.180 0.271820
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.002 on 8 degrees of freedom
Multiple R-squared:  0.9892, Adjusted R-squared:  0.9852
F-statistic: 244.8 on 3 and 8 DF,  p-value: 3.302e-08
```

Question: Find the least squares estimators of the regression coefficients and the regression line.
Solution:

$$\hat{\beta}_0 = 26.61393$$
$$\hat{\beta}_1 = 0.81677$$
$$\hat{\beta}_2 = -0.22854$$
$$\hat{\beta}_3 = 3.45283$$

The regression line is

$$\hat{y} = 26.61393 + 0.81677x_1 - 0.22854x_2 + 3.45283x_3$$

Question: Use the ANOVA table given by the R output to test the usefulness (or significance) of the full MLR model. Use $\alpha = 0.05$.

Step 1: State the hypotheses

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0 \quad \text{(all predictor variables together have no linear relationship with } y\text{)}$$
$$H_1 : \text{At least one } \beta_i \neq 0 \quad \text{(the model is useful/predictors significantly help explain } y\text{)}$$

Step 2: Identify the test statistic and its value From the R output:

$$\begin{aligned} F\text{-statistic} &= 244.8 \\ df_1 &= 3 \quad \text{(number of predictors)} \\ df_2 &= 8 \quad (n - k - 1 = 12 - 3 - 1 = 8) \\ p\text{-value} &= 3.302 \times 10^{-8} = 0.00000003302 \end{aligned}$$

Step 3: Decision at $\alpha = 0.05$ Since p-value $= 3.302 \times 10^{-8} < 0.05$, we **reject** H_0.

Step 4: Conclusion At the 5% significance level, there is sufficient evidence to conclude that at least one of the predictor variables (x_1, x_2, x_3) is useful for predicting y. In other words, the multiple linear regression model is statistically significant.

Question: Test whether x_2 contributes significantly in predicting y using t test. Use $\alpha = 0.05$.

Solution: **Step 1: State the hypotheses**

$$H_0 : \beta_2 = 0 \quad (x_2 \text{ does not significantly contribute to predicting } y)$$
$$H_a : \beta_2 \neq 0 \quad (x_2 \text{ does significantly contribute to predicting } y)$$

Step 2: Identify the test statistic and its value From the R output:

$$t = -2.433$$
$$p\text{-value} = 0.041032$$
$$df = 8$$

Step 3: Decision at $\alpha = 0.05$ Since p-value $= 0.041032 < 0.05$, we **reject** H_0.
Step 4: Conclusion At the 5% significance level, there is sufficient evidence to conclude that x_2 contributes significantly to predicting y, given that x_1 and x_3 are already in the model.

Question: Find the value of r^2 and adjusted r^2, and interpret their meanings in this problem.
From the R output:

$$r^2 = 0.9892$$
$$\text{Adjusted } r^2 = 0.9852$$

Coefficient of Determination ($r^2 = 0.9892$): Approximately 98.92% of the variation in the response variable y can be explained by the linear relationship with the predictor variables x_1, x_2, and x_3. This indicates an extremely strong fit, meaning the model explains almost all of the variability in the data.
Adjusted $r^2 = 0.9852$: After adjusting for the number of predictor variables in the model (3 predictors), approximately 98.52% of the variation in y is explained by the model. The adjusted r^2 is slightly lower than r^2, which is expected because it penalizes for adding extra predictors to the model. The small difference (0.004) between r^2 and adjusted r^2 suggests that all three predictors contribute meaningfully to the model, and there is no significant overfitting.

Question: Use partial F-test to determine whether x2 contribute significantly to the MLR mode. Below the output R-code.

```
Analysis of Variance Table

Response: y
Df Sum Sq Mean Sq F value    Pr(>F)
x1         1 714.12  714.12 460.131 4.892e-09 ***
x3         1  17.10   17.10  11.015  0.008956 **
Residuals  9  13.97    1.55
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Analysis of Variance Table

Response: y
Df Sum Sq Mean Sq  F value    Pr(>F)
x1         1 714.12  714.12 711.5590 4.197e-09 ***
x2         1  21.64   21.64  21.5592  0.001659 **
x3         1   1.40    1.40   1.3929  0.271820
Residuals  8   8.03    1.00
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Step 1: State the hypotheses

$$H_0 : \beta_2 = 0 \quad (x_2 \text{ does not significantly contribute to the model})$$
$$H_1 : \beta_2 \neq 0 \quad (x_2 \text{ does significantly contribute to the model})$$

Step 2: Extract the required values from the output

From the reduced model (without x_2):

$$SSE_{\text{reduced}} = 13.97, \quad df_{\text{reduced}} = 9$$

From the full model (with x_2):

$$SSE_{\text{full}} = 8.03, \quad df_{\text{full}} = 8$$

Step 3: Calculate the partial F-statistic

The partial F-statistic formula is:

$$F = \frac{(SSE_{\text{reduced}} - SSE_{\text{full}})/(df_{\text{reduced}} - df_{\text{full}})}{SSE_{\text{full}}/df_{\text{full}}}$$

Plugging in the values:

$$F = \frac{(13.97 - 8.03)/(9 - 8)}{8.03/8} = \frac{5.94/1}{1.00375} = 5.92$$

The numerator degrees of freedom is:

$$df_1 = df_{\text{reduced}} - df_{\text{full}} = 9 - 8 = 1$$

Step 4: Determine the critical value
At $\alpha = 0.05$, with $df_1 = 1$ and $df_2 = 8$, the critical value from the F-distribution table is:

$$F_{\text{critical}} = F_{0.05,1,8} = 5.32$$

Step 5: Decision rule Reject H_0 if $F > F_{\text{critical}}$.
Step 6: Compare and make a decision Since $F = 5.92 > F_{\text{critical}} = 5.32$, we **reject** H_0.
Step 7: Conclusion At the 5% significance level, there is sufficient evidence to conclude that x_2 contributes significantly to predicting y, given that x_1 and x_3 are already in the model.
Note The partial F-statistic ($F = 5.92$) is approximately equal to the square of the t-statistic for x_2 from the full model output:

$$t^2 = (-2.433)^2 = 5.92$$

This demonstrates the relationship between the partial F-test and the t-test for individual predictors in multiple linear regression.